Bracketing hyperoperations: a rank dichotomy for the Tamari order

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Abstract

Let H_r denote the Goodstein hyperoperation hierarchy, so that H_1 is addition, H_2 multiplication, H_3 exponentiation and H_4 tetration. Since H_r is non-associative for $r \geq 3$, the value of a bracketed expression $\mathrm{H}_r(a, \ldots, a)$ depends on the bracketing, i.e. on a binary tree. We study how the value varies over the Tamari lattice of such trees.

We prove that a *semi-associative law* runs through the whole hierarchy: for every rank $r \geq 3$ and all $x, y \geq 2$,

$$\mathrm{H}_r(\mathrm{H}_r(x, y), z) \;\leq\; \mathrm{H}_r(x, \mathrm{H}_r(y, z)),$$

so a right rotation at any node never decreases the value and evaluation is order-preserving from the Tamari lattice to $(\mathbb{N}, \leq)$; the left comb is a minimiser and the right comb, whose value is $\mathrm{H}_{r+1}(a, n)$, a maximiser.

Our main result is that the *degeneracy* of this law disappears at rank 4. We classify equality completely. For $x, y \geq 2$:

$$r = 3 : \quad \text{equality} \iff z = 1 \;\text{ or }\; (y, z) = (2, 2); \qquad r \geq 4 : \quad \text{equality} \iff z = 1.$$

Thus rank 3 carries exactly one non-trivial *equality pattern* — already visible for $n = 4$ leaves, where the Tamari cover $((aa)a)a \lessdot (aa)(aa)$ has equal endpoints — and from rank 4 on the evaluation is strictly increasing along *every* such cover (Theorem 1.4). (At rank 3 with $a = 2$ and $n = 4$ several trees share a value; what is unique is the non-trivial equality pattern $(y, z) = (2, 2)$ of the local law itself.) The rank-3 equality is produced by the collapsing law $(x^y)^z = x^{yz}$, which reduces the rank-3 inequality to $yz \leq y^z$; the classification shows that no equality of this kind survives at rank 4 or above.

The result is not an asymptotic triviality. The two sides compare a growing *first* argument against a growing *second* one, and small arguments do not blow up with the rank: $\mathrm{H}_r(2, 2) = 4$ for every $r \geq 2$. Moreover the hypothesis $y \geq 2$ is sharp as a uniform hypothesis: for every $r \geq 3$ and all $x, z \geq 2$, taking $y = 1$ reverses the inequality.

The monotonicity and strictness statements are proved for arbitrary leaf values ≥ 2, not only for a single repeated one, which extends to the whole hierarchy a rank-3 observation that appears to have been made only for exponentiation.

Every numbered statement of the paper that asserts an inequality or an equality about H_r or about the evaluation of trees is formalized and machine-checked in Lean 4; the classical structure of the Tamari lattice itself (that it is a lattice with the two combs as its extreme elements) is used as background and is not formalized. The development contains no `sorry`, every theorem is audited with `#print axioms`, and no declaration depends on `Classical.choice`.

1 Introduction

Exponentiation is not associative, and it is classical that among all bracketings of $a^{a^{\cdot^{\cdot^{\cdot^{a}}}}}$ with n occurrences of $a \geq 2$ the right-associated one is the largest; its value is the tetration $a \uparrow\uparrow n$. Orders on such bracketings have been discussed before — see Section 9 — but we are not aware of a treatment of the *Tamari* order on them, nor of the behaviour of that order further up the hyperoperation hierarchy.

Two strands of existing work bracket the question. On the one hand, the *associative spectrum* of a binary operation, introduced by Csákány and Waldhauser [1], measures non-associativity by counting how many distinct values the C_{n-1} bracketings can take; they prove in particular that exponentiation is *Catalan*, i.e. distinct bracketings induce distinct operations. That line of work counts values. On the other hand, the set of bracketings carries the Tamari lattice [12], generated by right rotations, whose proof-theoretic content was isolated by Zeilberger [13] as a semi-associative law. That line of work orders bracketings. The present paper connects the two: we show that for hyperoperations the Tamari order is *respected* by evaluation, and we locate precisely where the correspondence degenerates.

Results

Throughout, H_r is the hyperoperation hierarchy fixed in Section 2, with $\mathrm{H}_1(a,b) = a+b$, $\mathrm{H}_2(a,b) = ab$, $\mathrm{H}_3(a,b) = a^b$ and $\mathrm{H}_4(a,b) = a \uparrow\uparrow b$.

Theorem 1.1 (Rank 3, complete solution)**.** *Let $x \geq 1$ and $y \geq 2$. Then $(x^y)^z \leq x^{(y^z)}$ for every $z \geq 0$. If moreover $x \geq 2$, equality holds if and only if $z = 1$ or $(y,z) = (2,2)$. The hypothesis $y \geq 2$ cannot be dropped: for $y = 1$ and all $x, z \geq 2$ the inequality fails.*

Theorem 1.2 (Rank 4)**.** *Let $x \geq 2$ and $y \geq 2$. Then $(x \uparrow\uparrow y) \uparrow\uparrow z \leq x \uparrow\uparrow (y \uparrow\uparrow z)$ for every $z \geq 0$. If moreover $z \geq 2$, the inequality is strict.*

Proposition 1.3 (Tamari monotonicity)**.** *Let $a \geq 2$ and $r \geq 3$. If $T \lessdot U$ is a Tamari cover, i.e. U arises from T by a right rotation at some node, then $\mathrm{ev}_{r,a}(T) \leq \mathrm{ev}_{r,a}(U)$. Consequently $\mathrm{ev}_{r,a}$ is order-preserving for the Tamari order. Moreover, for the right comb R_n with n leaves, $\mathrm{ev}_{r,a}(R_n) = \mathrm{H}_{r+1}(a,n)$.*

Theorem 1.4 (Strict Tamari monotonicity from rank four)**.** *Let $a \geq 2$ and $r \geq 4$. If $T \lessdot U$ is a Tamari cover, then*

$$\mathrm{ev}_{r,a}(T) < \mathrm{ev}_{r,a}(U).$$

Thus for $r \geq 4$ the evaluation is strictly increasing along every *cover of the Tamari lattice, not merely along root rotations.*

Corollary 1.5 (Rank dichotomy)**.** *For $a \geq 2$ the evaluation $\mathrm{ev}_{r,a}$ is order-preserving for the Tamari order at every rank $r \geq 3$ (Proposition 1.3), and*

- *at rank 3 strictness can fail, but only for $a = 2$: the cover $((aa)a)a \lessdot (aa)(aa)$ carries equal values, both 256, when $a = 2$;*
- *at every rank $r \geq 4$ it is strict on covers (Theorem 1.4).*

In fact $(r,a) = (3,2)$ is the only exception: for every other pair with $r \geq 3$ and $a \geq 2$ every cover is strict (Corollary 7.3).

Theorem 1.6 (All ranks)**.** *For every $r \geq 3$ and all $x \geq 2$, $y \geq 2$ and $z \geq 0$,*

$$\mathrm{H}_r\big(\mathrm{H}_r(x,y),z\big) \;\leq\; \mathrm{H}_r\big(x,\mathrm{H}_r(y,z)\big).$$

Theorem 1.7 (Complete equality classification)**.** *Let $x \geq 2$ and $y \geq 2$.*

1. *For every $r \geq 2$ the two sides are equal when $z = 1$; in particular equality holds in Theorem 1.6 for every $r \geq 3$.*

2. For $r = 3$ the only further equality is $(y, z) = (2, 2)$.

3. For $r \geq 4$ and $z \geq 2$ the inequality is strict.

4. For every $r \geq 3$ the inequality is strict when $z = 0$, since $\mathrm{H}_r(\mathrm{H}_r(x, y), 0) = 1 < x = \mathrm{H}_r(x, \mathrm{H}_r(y, 0))$.

Items (1), (3) and (4) together give: for $r \geq 4$ equality holds if and only if $z = 1$.

Theorems 1.6 and 1.7 are proved in Section 6 by inductions on the rank whose engine is the identity-like inequality (SUM_r) below; the case $r = 3$ is Theorem 1.1 and the case $r = 4$ can also be obtained directly, as in Section 4.

Remark 1.8 (The statement is not an asymptotic triviality)**.** Two features are worth isolating. First, (HT_r) does not follow from monotonicity alone: on the left the *first* argument grows, $x \mapsto \mathrm{H}_r(x, y) > x$, while on the right the *second* argument grows, $z \mapsto \mathrm{H}_r(y, z)$, so the two coordinates compete. Second, high rank does not make everything large: since $\mathrm{H}_r(a, 1) = a$ for $r \geq 2$, one gets

$$\mathrm{H}_r(2, 2) = \mathrm{H}_{r-1}(2, \mathrm{H}_r(2, 1)) = \mathrm{H}_{r-1}(2, 2) = \cdots = \mathrm{H}_2(2, 2) = 4$$

for *every* $r \geq 2$. A degenerate corner therefore survives at every rank, and it is exactly this corner that carries the rank-3 equality of Theorem 1.7(2) and that has to be treated separately in the proof of Theorem 1.7(3).

Formalization

Every theorem stated here is proved in Lean 4 on top of Mathlib. The development defines the hyperoperation by the usual double recursion and *verifies the convention* by proving $\mathrm{H}_1 = +$, $\mathrm{H}_2 = \cdot$, $\mathrm{H}_3 = (\ )^{(\ )}$; this matters because the literature on hyperoperations uses several incompatible indexings and initial values. Section 8 describes the development.

2 Preliminaries

Definition 2.1 (Hyperoperation [7])**.** Define $\mathrm{H} : \mathbb{N} \times \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ by

$$\mathrm{H}_0(a, b) = b + 1, \quad \mathrm{H}_1(a, 0) = a, \quad \mathrm{H}_2(a, 0) = 0, \quad \mathrm{H}_{r+3}(a, 0) = 1,$$
$$\mathrm{H}_{r+1}(a, b + 1) = \mathrm{H}_r\bigl(a, \mathrm{H}_{r+1}(a, b)\bigr).$$

Lemma 2.2. *For all a, b we have $\mathrm{H}_1(a, b) = a+b$, $\mathrm{H}_2(a, b) = ab$ and $\mathrm{H}_3(a, b) = a^b$. Moreover $\mathrm{H}_r(a, 1) = a$ for $r \geq 2$.*

We write $a \uparrow\uparrow b := \mathrm{H}_4(a, b)$, so $a \uparrow\uparrow 0 = 1$ and $a \uparrow\uparrow (b + 1) = a^{(a\uparrow\uparrow b)}$.

Definition 2.3 (Bracketings and evaluation)**.** Let $\mathcal{T}_n$ denote the set of binary trees with n leaves. For $r, a \in \mathbb{N}$ define $\mathrm{ev}_{r,a} : \mathcal{T}_n \to \mathbb{N}$ by $\mathrm{ev}_{r,a}(\mathrm{leaf}) = a$ and $\mathrm{ev}_{r,a}(T \cdot U) = \mathrm{H}_r\bigl(\mathrm{ev}_{r,a}(T), \mathrm{ev}_{r,a}(U)\bigr)$.

Definition 2.4 (Right rotation)**.** The *right rotation at the root* sends $((A \cdot B) \cdot C)$ to $(A \cdot (B \cdot C))$. The Tamari order on $\mathcal{T}_n$ is generated by right rotations performed at arbitrary nodes; that this order is a lattice, with the left comb as least and the right comb as greatest element, is the theorem of Friedman and Tamari [2], whose title already names the relation a *loi demi-associative* [12, 13].

Thus the inequality

$$\mathrm{H}_r\bigl(\mathrm{H}_r(x, y), z\bigr) \ \leq\ \mathrm{H}_r\bigl(x, \mathrm{H}_r(y, z)\bigr) \qquad (\mathrm{HT}_r)$$

is exactly the assertion that a right rotation, applied with x, y, z the values of the three subtrees, does not decrease the value.

3 Rank three

At rank 3 the collapsing law $(x^y)^z = x^{yz}$ reduces (HT_r) to a comparison of exponents.

Lemma 3.1. *If $y \geq 2$ then $yz \leq y^z$ for every z. If moreover $z \geq 3$ the inequality is strict.*

Proposition 3.2. *If $y \geq 2$ then $yz = y^z$ if and only if $z = 1$ or $(y, z) = (2, 2)$.*

Proof. If $z = 0$ then $yz = 0 \neq 1 = y^z$. If $z = 1$ both sides are y. If $z = 2$ then $yz = y^z$ reads $2y = y^2$, i.e. $y = 2$ after cancelling $y > 0$. If $z \geq 3$ the inequality is strict by Lemma 3.1. The converse is immediate. □

Proof of Theorem 1.1. Since $(x^y)^z = x^{yz}$, the inequality follows from Lemma 3.1 and monotonicity of $t \mapsto x^t$ for $x \geq 1$. For $x \geq 2$ the map $t \mapsto x^t$ is injective, so equality holds iff $yz = y^z$, and Proposition 3.2 applies. For the last claim, $(x^1)^z = x^z$ while $x^{(1^z)} = x$, and $x^z > x$ whenever $x, z \geq 2$. □

Example 3.3 (Failure of strictness)**.** Take $a = 2$ and $n = 4$. The five bracketings evaluate under H_3 to

$$((aa)a)a = 256, \quad (a(aa))a = 256, \quad (aa)(aa) = 256, \quad a((aa)a) = 65536, \quad a(a(aa)) = 65536.$$

In particular the Tamari cover $((aa)a)a \lessdot (aa)(aa)$ has equal endpoints, so $\mathrm{ev}_{3,2}$ is monotone but not strictly monotone; it is also not injective.

4 Rank four

At rank 4 we know of no identity that would play the role of $(x^y)^z = x^{yz}$, and a different argument is required. The proof below is by induction on z; the base case $z = 1$ must be treated separately, since the inductive hypothesis at $z = 0$ is too weak to survive the step.

Lemma 4.1 (Squaring)**.** *If $x \geq 2$ and $m \geq 1$ then $(x \uparrow\uparrow m)^2 \leq x \uparrow\uparrow (m + 1)$.*

Proof. Induction on m. For $m = 1$ the claim is $x^2 \leq x^x$, which holds as $x \geq 2$. For the step put $t = x \uparrow\uparrow m$; then $(x \uparrow\uparrow (m + 1))^2 = (x^t)^2 = x^{2t} \leq x^{(x^t)} = x \uparrow\uparrow (m + 2)$, where $2t \leq 2^t \leq x^t$ uses $t \geq 1$ and $x \geq 2$. □

Remark 4.2. The direct inequality $t^2 \leq x^t$ is *false* in general — take $x = 2$, $t = 3$, where $t^2 = 9 > 8 = x^t$ — so the inductive formulation in Lemma 4.1 is essential.

Lemma 4.3. *If $y \geq 2$ and $m \geq 2$ then $m + 2 \leq y^m$. If moreover $m \geq 3$ then $m + 3 \leq y^m$.*

Proof of Theorem 1.2. Write $y = y' + 1$ with $y' \geq 1$ and induct on z. For $z = 0$ the left side is 1 and the right side is $x \geq 1$. For $z = 1$ both sides equal $x \uparrow\uparrow y$. Assume $z \geq 1$ and set $m := y \uparrow\uparrow z$, so $m \geq y \geq 2$ and $y' \leq m$. By Lemma 4.3 we have $y^m \geq m + 2$, so $y^m = k + 1$ with $k \geq m + 1$. Writing $A := x \uparrow\uparrow y = x^{(x \uparrow\uparrow y')}$ we compute

$$\begin{aligned}(x \uparrow\uparrow y) \uparrow\uparrow (z + 1) = A^{(x\uparrow\uparrow y)\uparrow\uparrow z} \;&\leq\; A^{x\uparrow\uparrow m} \;=\; \big(x^{x\uparrow\uparrow y'}\big)^{x\uparrow\uparrow m} \;=\; x^{(x\uparrow\uparrow y')(x\uparrow\uparrow m)} \\ &\leq x^{(x\uparrow\uparrow m)^2} \;\leq\; x^{x\uparrow\uparrow(m+1)} \;\leq\; x^{x\uparrow\uparrow k} \;=\; x \uparrow\uparrow (k + 1) \;=\; x \uparrow\uparrow \big(y \uparrow\uparrow (z + 1)\big),\end{aligned}$$

using the inductive hypothesis, $x \uparrow\uparrow y' \leq x \uparrow\uparrow m$, Lemma 4.1, and $m + 1 \leq k$. This proves the inequality.

For strictness, note that the penultimate step is strict as soon as $m \geq 3$: then $y^m \geq m+3$ by Lemma 4.3, hence $m + 1 < k$ and $x \uparrow\uparrow (m + 1) < x \uparrow\uparrow k$. Now $m = y \uparrow\uparrow z \geq 3$ holds whenever $z \geq 2$ (then $m \geq 4$), and also when $z = 1$ and $y \geq 3$ (then $m = y$). This gives strictness at $z+1$ for every $z+1 \geq 2$ except $(y, z+1) = (2,2)$, which we compute directly:

$$(x \uparrow\uparrow 2) \uparrow\uparrow 2 = (x^x)^{x^x} = x^{x \cdot x^x} = x^{x^{x+1}}, \qquad x \uparrow\uparrow (2 \uparrow\uparrow 2) = x \uparrow\uparrow 4 = x^{x^{x^x}},$$

and $x + 1 < x^x$ for $x \geq 2$. Hence the inequality is strict for every $z \geq 2$. □

5 The dichotomy

Proof of Corollary 1.5. Order-preservation at every rank $r \geq 3$ is Proposition 1.3. Non-strictness at rank 3 is Example 3.3. Strictness at rank 4 is Theorem 1.2; for general $r \geq 4$ it is Theorem 1.4, which is proved in Section 6 and rests on Theorem 1.7(3). (The present section proves only the rank-4 case, so that the dichotomy is available before the general induction.) □

Proof of Proposition 1.3. Induction on the derivation of the cover. At the root the claim is Theorem 1.6 instantiated at the values of the three subtrees, which are all ≥ 2 because $a \geq 2$. If the rotation takes place inside the left subtree, one uses monotonicity of $\mathrm{H}_r(\cdot, v)$ in its first argument; inside the right subtree, monotonicity of $\mathrm{H}_r(u, \cdot)$ in its second argument, valid for $u \geq 2$. Both monotonicity statements hold for every $r \geq 1$ and are proved by a double induction on the rank and on the argument. The formula for the right comb is immediate from $\mathrm{H}_{r+1}(a, n+1) = \mathrm{H}_r(a, \mathrm{H}_{r+1}(a, n))$ together with $\mathrm{H}_{r+1}(a, 1) = a$. □

The mechanism is worth stating plainly. At rank 3 the identity $(x^y)^z = x^{yz}$ turns (HT_r) into $yz \leq y^z$, an inequality which is *tight*: it becomes an equality at $(y, z) = (2, 2)$. Each of the three coincidences in Example 3.3 is an instance of this one equality, applied at a node whose two lower subtrees both evaluate to 2: the cover $((aa)a)a \lessdot (aa)(aa)$ is the root rotation with $(x, y, z) = (4, 2, 2)$, and the pairs $((aa)a)a$, $(a(aa))a$ and $a((aa)a)$, $a(a(aa))$ differ only in a subtree $(aa)a = a(aa) = 16$, which is the root rotation with $(x, y, z) = (2, 2, 2)$. At rank 4 Theorem 1.2 shows that no equality with $z \geq 2$ occurs at all, and the gap is already visible at the smallest instance: for $x = y = z = 2$ the two sides are 256 and 65536.

6 General rank

We now prove Theorem 1.6. The engine is the following inequality; write

$$\mathrm{H}_{r-1}\big(\mathrm{H}_r(x, a), \mathrm{H}_r(x, b)\big) \ \leq\ \mathrm{H}_r(x,\ a + b). \qquad (\mathrm{SUM}_r)$$

Proposition 6.1. *Let $r \geq 3$ and $x \geq 2$.*

1. *If (HT_{r-1}) holds, then (SUM_r) holds for all $a, b \geq 1$.*
2. *If (SUM_r) holds and $u + v \leq \mathrm{H}_{r-1}(u, v)$ for all $u, v \geq 2$, then (HT_r) holds for all $y \geq 2$.*

Proof. (1) Induct on a. For $a = 1$ both sides are equal:

$$\mathrm{H}_{r-1}\big(\mathrm{H}_r(x, 1), \mathrm{H}_r(x, b)\big) = \mathrm{H}_{r-1}\big(x, \mathrm{H}_r(x, b)\big) = \mathrm{H}_r(x, b+1).$$

For the step, $\mathrm{H}_r(x, a+1) = \mathrm{H}_{r-1}(x, \mathrm{H}_r(x, a))$, so by (HT_{r-1})

$$\mathrm{H}_{r-1}\big(\mathrm{H}_{r-1}(x, \mathrm{H}_r(x, a)), \mathrm{H}_r(x, b)\big) \leq \mathrm{H}_{r-1}\big(x, \mathrm{H}_{r-1}(\mathrm{H}_r(x, a), \mathrm{H}_r(x, b))\big),$$

and the inductive hypothesis bounds the inner term by $\mathrm{H}_r(x, a+b)$, giving $\mathrm{H}_{r-1}(x, \mathrm{H}_r(x, a+b)) = \mathrm{H}_r(x, a+b+1)$.

(2) Induct on z, with $z = 0, 1$ as in Theorem 1.2. For the step set $m := \mathrm{H}_r(y, z) \geq 2$; then $\mathrm{H}_r(\mathrm{H}_r(x, y), z+1) = \mathrm{H}_{r-1}(\mathrm{H}_r(x, y), \mathrm{H}_r(\mathrm{H}_r(x, y), z)) \leq \mathrm{H}_{r-1}(\mathrm{H}_r(x, y), \mathrm{H}_r(x, m)) \leq \mathrm{H}_r(x, y+m) \leq \mathrm{H}_r(x, \mathrm{H}_{r-1}(y, m)) = \mathrm{H}_r(x, \mathrm{H}_r(y, z+1))$. ☐

Proof of Theorem 1.6. Induction on r. The base case $r = 3$ is Theorem 1.1. For the step, the inductive hypothesis is exactly (HT_{r-1}), so Proposition 6.1(1) gives (SUM_r), and Proposition 6.1(2) then gives (HT_r), the hypothesis $u + v \leq \mathrm{H}_{r-1}(u, v)$ for $u, v \geq 2$ being Lemma 6.2. ☐

Lemma 6.2. *For $u, v \geq 2$ and $s \geq 2$ we have $u + v \leq \mathrm{H}_s(u, v)$.*

Proof. $u + v \leq uv = \mathrm{H}_2(u, v)$, and $\mathrm{H}_2(u, v) \leq \mathrm{H}_s(u, v)$ by monotonicity in the rank. ☐

Strictness

Lemma 6.3. *For $u, v \geq 2$ with $u \geq 3$ or $v \geq 3$, and $s \geq 2$, we have $u + v < \mathrm{H}_s(u, v)$. Equality $u + v = \mathrm{H}_s(u, v)$ occurs exactly at $u = v = 2$, where both sides are 4.*

Proof. $u + v < uv$ holds iff $(u-1)(v-1) > 1$, i.e. iff $u \geq 3$ or $v \geq 3$; and $uv = \mathrm{H}_2(u, v) \leq \mathrm{H}_s(u, v)$. At $u = v = 2$ we have $u + v = 4 = \mathrm{H}_2(2, 2) = \mathrm{H}_s(2, 2)$ by Remark 1.8. ☐

Proof of Theorem 1.7. (1) $\mathrm{H}_r(\mathrm{H}_r(x, y), 1) = \mathrm{H}_r(x, y) = \mathrm{H}_r(x, \mathrm{H}_r(y, 1))$ since $\mathrm{H}_r(a, 1) = a$. (2) is Theorem 1.1. For (3) we induct on $r \geq 4$, the base $r = 4$ being the strictness clause of Theorem 1.2. Let $r \geq 5$ and write $z = w + 1$ with $w \geq 1$, and $m := \mathrm{H}_r(y, w) \geq 2$.

If $y \geq 3$ or $w \geq 2$, then $y \geq 3$ or $m \geq 4$, so Lemma 6.3 makes the last step of the proof of Proposition 6.1(2) strict, and strictness propagates because $\mathrm{H}_r(x, \cdot)$ is strictly increasing.

The remaining case is $y = z = 2$. Put $A := \mathrm{H}_r(x, 2) = \mathrm{H}_{r-1}(x, x) \geq 4$. Then

$$\mathrm{H}_r(\mathrm{H}_r(x, 2), 2) = \mathrm{H}_{r-1}(A, A), \qquad \mathrm{H}_r(x, \mathrm{H}_r(2, 2)) = \mathrm{H}_r(x, 4) = \mathrm{H}_{r-1}\big(x, \mathrm{H}_{r-1}(x, A)\big),$$

using $\mathrm{H}_r(2, 2) = 4$ from Remark 1.8. Since $A = \mathrm{H}_{r-1}(x, x)$, the required inequality

$$\mathrm{H}_{r-1}\big(\mathrm{H}_{r-1}(x, x), A\big) < \mathrm{H}_{r-1}\big(x, \mathrm{H}_{r-1}(x, A)\big)$$

is the inductive hypothesis at rank $r - 1$ applied to (x, x, A), and $A \geq 2$. ☐

We can now upgrade Proposition 1.3 to its strict form.

Proof of Theorem 1.4. Induction on the derivation of the cover, exactly as for Proposition 1.3 but with every inequality replaced by its strict counterpart. At the root the claim is Theorem 1.7(3) instantiated at the values of the three subtrees, which are all ≥ 2 because $a \geq 2$ (so in particular the hypothesis $z \geq 2$ of that item is met). If the rotation takes place inside the left subtree, one needs

$$2 \leq u < u' \implies \mathrm{H}_r(u, v) < \mathrm{H}_r(u', v) \qquad (v \geq 1),$$

the strict form of monotonicity in the first argument; if it takes place inside the right subtree, one needs

$$1 \leq v < v' \implies \mathrm{H}_r(u, v) < \mathrm{H}_r(u, v') \qquad (u \geq 2,\ r \geq 2),$$

the strict form of monotonicity in the second argument. Both are proved by the same double induction on the rank and on the argument as their non-strict counterparts, and both apply here because every subtree value is ≥ 2. ☐

The auxiliary facts used above — monotonicity of $\mathrm{H}_r(x,\cdot)$ in its second argument, monotonicity in the rank, and $n+1 \le \mathrm{H}_r(x,n)$ for $x \ge 2$, $r \ge 2$, $n \ge 1$ — are elementary and are proved in the formalization by a sequence of inductions.

The hypothesis $y \ge 2$ of Theorem 1.6 is sharp, in the following precise sense: for every $r \ge 3$ and all $x, z \ge 2$, taking $y = 1$ reverses the inequality. Indeed $\mathrm{H}_r(x,1) = x$, so the left side is $\mathrm{H}_r(x,z)$, whereas $\mathrm{H}_r(1,z) = 1$ makes the right side $\mathrm{H}_r(x,1) = x$, and $\mathrm{H}_r(x,z) > x$ because $z \ge 2$. The restriction to $x, z \ge 2$ is necessary and not an artefact: for $x = 1$ both sides equal 1, since $\mathrm{H}_r(1,\cdot) = 1$; for $z = 1$ they are equal by Theorem 1.7(1); and for $z = 0$ the left side is 1 and the right side is x, so the original inequality still holds. For $x = 0$ the operations degenerate.

7 Arbitrary leaf values

Nothing so far has used the fact that the leaves all carry the same value. The local law (HT_r), and its strict form, are statements about three independent arguments; only the evaluation $\mathrm{ev}_{r,a}$ was tied to a single label. Removing that restriction costs nothing and gives a strictly stronger statement, which is moreover the one that the discussion recalled in Section 9 was about.

Definition 7.1 (Labelled trees)**.** A *labelled tree* is a binary tree each of whose leaves carries a natural number. Its evaluation at rank r is defined by $\mathrm{ev}_r(\ell) = \ell$ for a leaf labelled ℓ and $\mathrm{ev}_r(T \cdot U) = \mathrm{H}_r(\mathrm{ev}_r T, \mathrm{ev}_r U)$. Right rotation at a node is defined exactly as before; it permutes no labels.

Theorem 7.2 (Arbitrary leaf values)**.** *Let T be a labelled tree all of whose leaf labels are ≥ 2, and let $T \lessdot U$ be a Tamari cover. Then*

$$\mathrm{ev}_r(T) \le \mathrm{ev}_r(U) \quad (r \ge 3), \qquad \mathrm{ev}_r(T) < \mathrm{ev}_r(U) \quad (r \ge 4),$$

and the same holds along the Tamari order itself, that is along the reflexive transitive closure of the covering relation in the first case and its transitive closure in the second.

Proof. Rotation does not change the multiset of leaf labels, so the hypothesis is preserved along a cover, and every subtree of T evaluates to at least 2 at every rank $r \ge 2$ by induction on the tree. The induction on the derivation of the cover is then the one already used for Proposition 1.3 and Theorem 1.4: at the root the claim is Theorem 1.6 respectively Theorem 1.7(3), instantiated at the three subtree values, which are independent of one another and all ≥ 2; inside the left subtree one uses monotonicity, respectively strict monotonicity, of $\mathrm{H}_r(\cdot, v)$ in its first argument, and inside the right subtree the corresponding statement for $\mathrm{H}_r(u, \cdot)$. The passage to the closures is by induction on the length of the chain. $\square$

We can now say exactly when a cover fails to be strict. At rank 3 the only non-trivial equality of (HT_r) is $(y,z) = (2,2)$, and in a labelled tree whose labels are all ≥ 2 an *internal* node evaluates to at least $\mathrm{H}_r(2,2) = 4$; so a subtree evaluates to 2 exactly when it is a leaf labelled 2.

Corollary 7.3 (Equality along a cover)**.** *Let all labels be ≥ 2 and consider the rotation $((A \cdot B) \cdot C) \to (A \cdot (B \cdot C))$ at rank 3. Its two sides are equal if and only if B and C are both leaves labelled 2. Consequently, for a single repeated label a, every cover is strict unless $(r,a) = (3,2)$; and for $(r,a) = (3,2)$ equality does occur, as Example 3.3 shows.*

Proof. By Theorem 1.1 the two sides agree exactly when $\mathrm{ev}(C) = 1$ or $(\mathrm{ev}(B), \mathrm{ev}(C)) = (2,2)$. The first is impossible because every label is ≥ 2. For the second, a subtree evaluates to 2 only if it is a leaf labelled 2, by the remark above. For a single label a: if $a \ge 3$ no leaf is labelled 2, so no cover at rank 3 is an equality, and from rank 4 on Theorem 1.4 applies. $\square$

Thus the collapse at rank 3 is not merely rare but completely located: it happens at exactly one local configuration, two leaves labelled 2 being rotated past each other, and from rank 4 on it does not happen at all.

Setting every label equal to a recovers Proposition 1.3 and Theorem 1.4; the formalization carries out this specialisation explicitly, so the earlier statements are corollaries rather than separate results.

Theorem 7.2 is what extends the rank-3 discussion of Section 9 to the whole hierarchy. We emphasise again what it does not do: it does not decide when two incomparable bracketings can be separated in both directions, which is the half that was left open there.

8 The Lean development

The development is written for Lean 4 (v4.30.0) with Mathlib (v4.30.0) and is available at

`https://github.com/turahigashi/hyper-tamari`

It consists of four files, `HyperTamari/Basic.lean`, `HyperTamari/Tetration.lean`, `HyperTamari/General.le` and `HyperTamari/VarLeaves.lean`, together with the raw `lake build` output recording the axiom audit (`logs/axiom-audit.txt`).

The correspondence with the numbered statements of this paper is given in Table 1. The population of the table is fixed as follows: it consists of *every* numbered environment in Sections 1–7 — that is, every `theorem`, `proposition`, `lemma`, `corollary`, `definition`, `example` and `remark` environment in the source of those sections. There are 24 of them: 6 theorems, 3 propositions, 6 lemmas, 2 corollaries, 4 definitions, 1 example and 2 remarks. (Section 8 contains no numbered environments.) Every one of the 24 appears in the table, and the only other row is one recording the specialisation of the labelled-tree results to a single repeated label; both facts, together with the existence in the development of every declaration name printed in the table, are checked by a script run on the source of this paper and of the Lean files. A numbered statement with several clauses may occupy several rows. We distinguish three cases, because they carry different evidential weight:

EXACT a single declaration with the same quantifiers, hypotheses and conclusion as the printed clause;

COLLECTIVE the clause is covered by a combination of declarations (typically because the paper states several things at once);

PROSE background material that is *not* formalized here — standard facts about the Tamari lattice that we use as context, or explanatory remarks.

We stress the third row of this classification: the paper does *not* claim that everything it says is machine-checked. What is machine-checked is every clause of every numbered statement that asserts an inequality or an equality about H_r or about the evaluation of trees; the table says which declaration carries which clause.

Table 1: Correspondence between the 24 numbered statements of Sections 1–7 and the Lean development. Clauses of one statement are listed on separate rows.

Statement	Lean declaration(s)	Status
Thm. 1.1 (inequality)	`ht_rank3`	EXACT
Thm. 1.1 (equality iff)	`ht_rank3_eq_iff`	EXACT

Table 1 (continued)

Statement	Lean declaration(s)	Status
Thm. 1.1 (sharpness of $y \geq 2$)	`ht_reversed_at_y_one` at $r = 3$ (universal), `ht_needs_two_le_y` (the witness $x = z = 3$)	COLLECTIVE
Thm. 1.2 (inequality)	`tet_ht`, `ht_rank4`	EXACT
Thm. 1.2 (strict for $z \geq 2$)	`tet_ht_strict_of_two_le`, `ht_rank4_strict_two`	EXACT
Prop. 1.3 (covers)	`Rot.eval_le`	EXACT
Prop. 1.3 (Tamari order)	`Rot.eval_le_of_reflTransGen`	EXACT
Prop. 1.3 (right comb)	`rightComb`, `eval_rightComb`	EXACT
Thm. 1.4 (covers)	`Rot.eval_lt`	EXACT
Thm. 1.4 (Tamari order)	`Rot.eval_lt_of_transGen`	EXACT
Cor. 1.5	`Rot.eval_le`, `rank3_rotation_not_strict`, `Rot.eval_lt`	COLLECTIVE
Thm. 1.6	`ht_general`	EXACT
Thm. 1.7 (1)–(4)	`equality_classification`, whose four components are `ht_eq_at_one`, `ht_rank3_equality_witness` with `ht_rank3_eq_iff` for (2), `ht_strict_general`, `ht_lt_at_zero`	COLLECTIVE
Rem. 1.8 ($\mathrm{H}_r(x, y) > x$; $\mathrm{H}_r(2, 2) = 4$)	`self_lt_hyper`, `hyper_two_two`	COLLECTIVE
Rem. 1.8 (the two coordinates compete)	— (explanatory)	PROSE
Def. 7.1	`LTree`, `LTree.eval`, `LTree.AllLeavesTwoLe`, `LRot`	EXACT
Thm. 7.2	`LRot.eval_le`, `LRot.eval_lt`, `LRot.eval_le_of_reflTransGen`, `LRot.eval_lt_of_transGen`, `LRot.preserves`, `LTree.two_le_eval`	COLLECTIVE
Specialisation to a single label	`ofBTree`, `eval_ofBTree`, `rot_ofBTree`, `Rot.eval_lt_of_var`	EXACT
Cor. 7.3	`LTree.four_le_eval_node`, `LTree.eval_eq_two_iff`, `LRot.root_eq_rank3_iff`, `LRot.root_lt_rank3_of_three_le`, `rank3_root_lt_of_three_le`, `cover_strict_unless_rank3_two`	COLLECTIVE
Def. 2.1	`hyper`, `hyper_zero`, `hyper_one_zero`, `hyper_two_zero`, `hyper_succ_succ_zero`, `hyper_succ`	EXACT
Lem. 2.2	`hyper_one`, `hyper_two`, `hyper_three`, `hyper_one_arg`	COLLECTIVE
Def. 2.3 (trees, ev)	`BTree`, `BTree.eval` (not restricted to a fixed number of leaves)	EXACT
Def. 2.4 (rotation, Tamari order)	`Rot` (cover), `Relation.ReflTransGen Rot` (the order)	EXACT
Def. 2.4 ($\mathcal{T}_n$, lattice, comb extremality)	— (classical [2, 12], used as background)	PROSE
Lem. 3.1	`mul_le_pow_of_two_le`, `mul_lt_pow_of_three_le`	COLLECTIVE
Prop. 3.2	`mul_eq_pow_iff`	EXACT
Ex. 3.3 (the five values; the colliding cover)	`rank3_five_values`, `rank3_rotation_not_strict`	COLLECTIVE
Lem. 4.1	`sq_tet_le`	EXACT
Rem. 4.2	`sq_le_pow_fails` (the negated universal statement), `two_pow_three_lt_three_sq`	EXACT

Table 1 (continued)

Statement	Lean declaration(s)	Status
Lem. 4.3	`add_two_le_pow`, `add_three_le_pow`	COLLECTIVE
Prop. 6.1 (1), (2)	`sum_lemma`, `ht_of_sum`	COLLECTIVE
Lem. 6.2	`add_le_hyper`	EXACT
Lem. 6.3 (strict inequality; equality iff $u = v = 2$; value 4)	`add_lt_hyper`, `add_eq_hyper_iff`, `hyper_two_two`	COLLECTIVE

Two assertions of Section 6 are not numbered and therefore lie outside the population of the table; both are formalized: the sharpness statement for all ranks ("for every $r \geq 3$ and all $x, z \geq 2$, taking $y = 1$ reverses the inequality") is `ht_reversed_at_y_one`, and the equality of both sides at $x = 1$ is `ht_eq_at_x_one`. The value pair $256 < 65536$ quoted in Section 5 is `ht_rank4_strict_smallest`. The development also contains `pow_lt_pow_right'` and `pow_right_injective'`, choice-free replacements for the corresponding Mathlib lemmas (see below), and the auxiliary monotonicity facts `succ_le_hyper`, `hyper_mono_arg`, `hyper_mono_base`, `hyper_rank_mono`, `hyper_lt_arg` and `hyper_lt_base`.

Draft note — not part of the submission. The artifact has *not* yet been deposited in a long-term archive, so no DOI is quoted above. A version DOI will be minted and inserted here at the moment this paper is submitted, and the archive record will be linked to the submission. Until then the repository URL is the only pointer.

No declaration uses `sorry`, `native_decide` or any private axiom. Every one of the **111** theorems in these four files is audited with `#print axioms`, and the audit is machine-checked to cover them all (declared 111, audited 111, missing 0; the script is `scripts/check_audit.py`); `sorryAx` does not occur. The axiom dependencies are `propext` alone (5 theorems), `propext, Quot.sound` (102), and four theorems that depend on no axioms at all. In particular

no declaration depends on `Classical.choice`.

This is not a foundational claim, and choice-freeness is not what the paper is about: every statement here is an elementary assertion about natural numbers, and nobody would expect it to need choice. We record it because it costs nothing and makes the artifact reusable — a development free of `Classical.choice` can be transported into constructive settings, or used inside a larger development that must itself avoid choice, without reproving anything.

It is worth saying where the choice had been hiding, since it was not in the mathematics. An earlier version of this development did depend on `Classical.choice` in 30 of its declarations, and the dependency was traced to exactly four sources, all of them incidental: the Mathlib lemmas `Nat.pow_lt_pow_right` and `Nat.pow_right_injective` (whose companion `Nat.pow_le_pow_right` is proved with no axioms at all), the `nlinarith` tactic, and — the one that is easiest to miss — `norm_num` applied to a numeric *inequality*, which pulls in choice, while `omega` discharges the same goal without it and `norm_num` applied to an *equation* does not. Replacing these by `omega`, by `decide`, by `Nat.add_le_mul`, and by two short inductions built on the axiom-free `Nat.pow_lt_pow_succ` removed the dependency entirely, with no change to any statement.

9 Related work

Csákány and Waldhauser [1] introduce the associative spectrum and show that exponentiation is Catalan: for pairwise distinct prime leaves, distinct bracketings give distinct values. Their concern is the number of values; the order in which those values arrange themselves is not addressed, and the Tamari lattice appears in their bibliography only as the source of the enumeration of bracketings. Huang and Lehtonen [9] extend the invariant to the *associative-commutative spectrum* and determine it for exponentiation (their Proposition 6.1.5): the associative spectrum of exponentiation is C_{n-1} and its ac-spectrum is n^{n-1}. Their question is again how many distinct term operations the bracketings produce; ours is how the resulting values are *arranged*, and neither the Tamari order nor any monotonicity statement occurs in that line of work. Huang, Mickey and Xu [10] compute the associative spectrum of a further operation. Tamari [12] enumerated the bracketings, and Friedman and Tamari [2] proved that the right-rotation order on them is a finite lattice — their title already calls the relation a *loi demi-associative*, which is precisely the law we show to run through the whole hyperoperation hierarchy. Zeilberger [13] gives a sequent calculus for that law. Gryszka [8] studies the comparison of power towers by normalizing them, which concerns right-associated towers rather than general bracketings.

Where the rank-three case comes from, and what we add

It is important to say clearly what part of Proposition 1.3 is already known. In a MathOverflow question of January 2018 [3], Spivak treats exponentiation on the integers ≥ 2 as an ordered magma and orders bracketings by the *evaluation order*: $E \leq F$ when the inequality holds *for every* assignment of values to the leaves. He asks how this order relates to the height (Stanley) order on Dyck paths. In the answers, Chow proves that a Stanley cover does decrease the value, the computation resting on

$$B^{CD} = (B^C)^D \;\geq\; (BC)^D \;\geq\; BC^D \qquad (B, C \geq 2,\; D \geq 1),$$

and Rubey reduces the same covering relation to the inequality $L_2^{(L_1^{xR})} \geq (L_2^{L_1})^{(x^R)}$, which is exactly (HT_r) at rank 3 in a variable-leaf form.

Since the evaluation order quantifies over *all* leaf assignments it is stronger than the single-label evaluation $\mathrm{ev}_{3,a}$, and since the Stanley lattice is an extension of the Tamari lattice, comparability in the Tamari order implies comparability in the Stanley order. Consequently **the rank-3 monotonicity is already contained, in a stronger form, in that discussion, and we do not claim it.**

That discussion is, however, explicitly unfinished, and it is confined to rank 3. Chow states that he can "complete half of the proof", the half we have just quoted; the converse half — that incomparable bracketings can be separated in both directions by a suitable choice of leaf values — he leaves open, writing that he has "not quite figured out how to say this rigorously". Rubey likewise reduces the covering relation to the displayed inequality and stops there. Neither treats any rank above 3, and neither asks when equality holds.

Theorem 7.2 below takes up the first half at every rank. It says that for arbitrary leaf values ≥ 2 the evaluation is monotone along the Tamari order at every rank $r \geq 3$, and *strictly* monotone from rank 4 on. We do not settle Chow's converse half, which remains open.

There is also a body of peer-reviewed work on orders attached to iterated exponentials, and it is worth saying precisely how it relates to the present question, because the titles suggest a closer overlap than there is. Brunson [4] fixes $e \leq x_1 \leq \cdots \leq x_n$ and, *for each permutation* of the x_i, forms the tower $T(y, z, \ldots, w) = y^{z^{\cdots^{w}}}$, in which "association is always to the upper right"; the resulting partial order lives on the symmetric group S_n. Stembridge [11] and Griggs–Wachs [6] compare that order with the Bruhat order on S_n, the latter exhibiting a counterexample at $n = 5$ to their coincidence. Griggs [5] studies the poset of towers over the two-letter alphabet $\{a, b\}^n$, the tower again being "evaluated in the conventional way, from top down".

In every one of these the *bracketing is fixed* — it is the right comb — and what varies is the assignment of entries to its leaves. Here the situation is exactly the reverse: the entries are fixed, all equal to a, and what varies is the bracketing. The two families of questions are therefore orthogonal, and none of these papers contains the inequality studied here. (None of [4, 5, 6] contains an occurrence of the words *bracketing*, *Tamari*, *binary tree*, *Catalan*, *rotation* or *Dyck*.)

What we do claim

Not finding a result is not a proof that it is new, so we state the claim in pieces. The rank-3 monotonicity is known (above). For the following we have not found a prior occurrence: the local semi-associative law (HT_r) for *every* rank $r \geq 3$ of the Goodstein hierarchy; the complete classification of its equality cases, with the isolated rank-3 exception $(y, z) = (2, 2)$; the resulting dichotomy, strict on every Tamari cover from rank 4 on; and the machine-checked formalization.

Acknowledgements

Use of AI assistance

Artificial-intelligence assistants were used substantially in the preparation of this work, and the author sets that use out in detail here. The question studied was formulated by the author. Large language models were then used to search for the shape of the proofs, to write the whole of the Lean 4 development, to draft this text, and to check the literature; further language models were used as adversarial readers, and their criticism led to the present form of the main theorem, in particular to the classification of the equality cases in Theorem 1.7 and to the extension to arbitrary leaf values in Section 7. No AI system is an author. The overall direction of the work and the choice of what to claim are the author's, and the author is responsible for all of the submitted content.

Draft note — not part of the submission. The journal's policy requires that, where AI has assisted with a proof or an argument, the author verify every step personally and also verify every citation and attribution. When that has been done, the following sentence belongs here, and it must not be included before then: *The author has independently verified every step of the mathematical arguments printed in this paper, and every citation and attribution.* The bibliographic data have already been checked against the publishers' records; what remains is to read the three items still marked † in the original, and to check the printed arguments by hand.

Two independent checks stand behind the mathematical content. Every clause of every numbered statement of Sections 1–7 that asserts an inequality or an equality about H_r or about the evaluation of trees has been formalised and checked by the Lean 4 kernel; Table 1 says, clause by clause, which declaration carries it, and marks the places where the paper relies instead on classical background about the Tamari lattice or on explanatory prose. The informal proofs printed in the text are the author's rendering of the formal ones and are not themselves machine-checked; they are set out in full so that a reader can follow them without consulting the Lean sources.

What rests on judgement rather than on either check, and is stated here so that it can be examined, is the choice of definitions, the claim of novelty, the attribution of prior work, and the PROSE rows of Table 1.

Draft note — not part of the submission. Items marked † in the list below have *not* been consulted in the original at the time of writing; the bibliographic data were taken from secondary sources and independently cross-checked against Crossref, but the contents were not read. They are: **[FT]** Friedman–Tamari 1967, cited only for the origin of the lattice and of the term *loi demi-associative* (the phrase occurs in the title itself); **[Gry]** — *now consulted in the original* (Lith. Math. J. **62**(2), 192–206, 15 pages): it compares right-associated power towers with varying entries and heights and does not mention bracketings, binary trees, Catalan structures, rotations or any lattice, which confirms the description given in Section 9; **[Tamari]** Tamari 1962, cited only for the enumeration of bracketings and for the two combs being the extreme elements; and **[Ste]** Stembridge 1989, on Brunson's order, cited only to place the question among its peer-reviewed neighbours. **[Bru]**, **[Gri]** and **[GW]** were also identified late, in response to a referee report, but have since been consulted in the original and are no longer marked: all three fix the bracketing to the right comb and vary the entries, as now described in Section 9. None of the results of the present paper depends on these three items: they are cited for attribution and context, and every mathematical statement used in the proofs is either proved here or formalised in the accompanying Lean development. Before submission the three still-unchecked items **[FT]**, **[Tamari]** and **[Ste]** must be checked in the original, and this note removed by setting `\draftfalse`.

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